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Project

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BACKGROUND

Background

ICU

https://commons.wikimedia.org/wiki/File:Respiratory_therapist.jpg

- Patient monitoring
  - Plethora of devices
- False alarms (up to 80%)
  - Loss of time (35% of time responding to alarms)
  - Burden on the practitioners (noise, stress)
  - Alarm fatigue

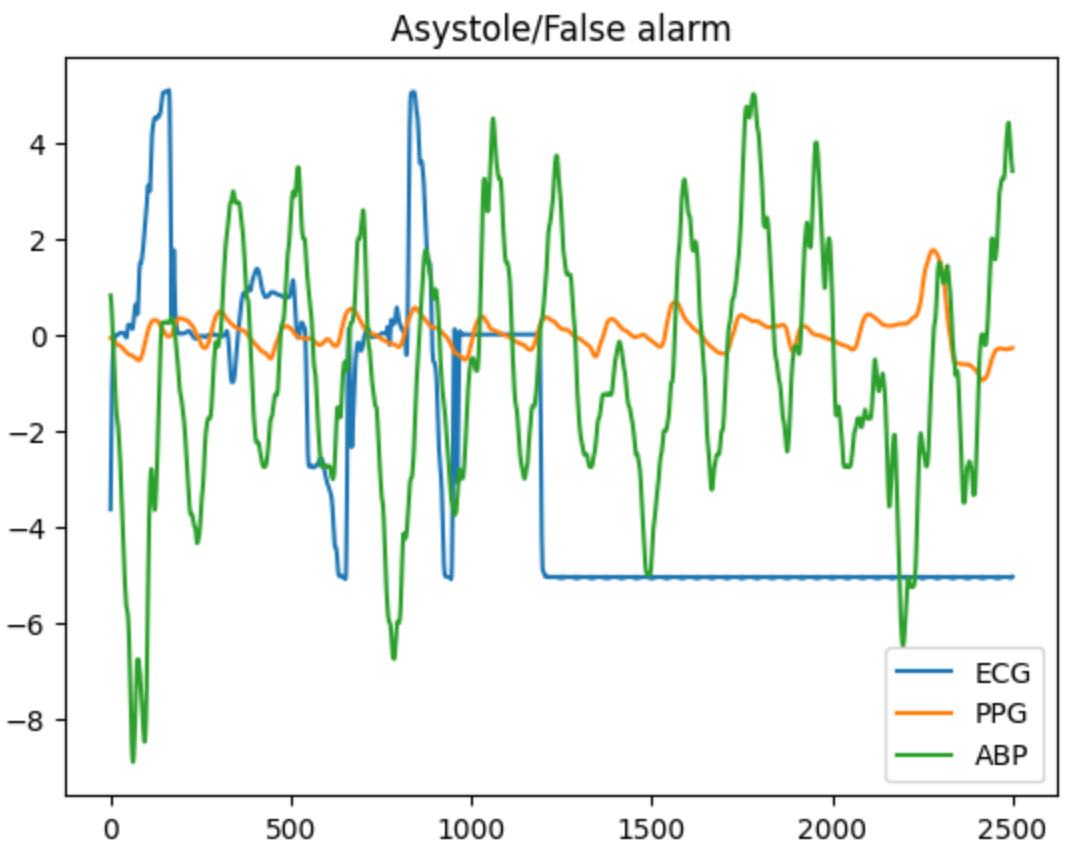
Lewandowska, K et. al. Impact of Alarm Fatigue on the Work of Nurses in an Intensive Care Environment—A Systematic Review. *Int. J. Environ. Res. Public Health* **2020**, *17*, 8409

02

DATASET

Dataset

Computing in Cardiology 2015 challenge



- Dataset based on the challenge
- 5 min recordings
  - ECG: different leads
  - PPG
  - ABP
  - 1 to 4 channels available
- 1 event within the last 10 seconds of the recordings triggers an alarm
- No information is provided about previous alarms within the recording.

Link to the dataset:
https://mediastore.rz.uni-augsburg.de/get/zNPE4V6i0G/

Dataset

Format

- Samples contain:
  - Sampling frequency
  - Channels of the recording
  - Type of alarm that was triggered
  - False / True alarm : your target
- As we splitted train and test set based on the dataset

  https://www.physionet.org/content/challenge-2015/1.0.0/

  Training based on the datasets version in the above link is therefore forbidden to prevent data leakage
- But you are allowed and encouraged to use other datasets if you have ideas of pretraining.

Event	Description
Asystole	No QRS at least 4 seconds
Bradycardia	Heartrate <40 bpm for at least 5 consecutive heartbeats
Tachycardia	Heartrate >140 bpm for at least 17 consecutive beats
Ventricular Tachycardia	At least 5 ventricular beats with heart rate >100 bpm
Ventricular Flutter	Fibrillatory, flutter, or oscillatory waveform for at least 4 seconds

03

TASK

Task

What you have to do

- Develop a deep learning based solution that:
  - Given an input 5 min recording containing ECG,PPG, ABP (or any of those channels), tells whether the alarm that was triggered within the last 10 seconds of the recording was **True or False**
- Submit your source code
  - The source code is based on the project zip provided on Digicampus
  - Your source code should train your deep learning model and save the model‘s parameters
  - We run your code to train your model on the provided train set (train_model.py)
  - We run your code to test your model a hidden test set (run_model.py)
  - **Warning: pay attention to train_model.py and run_model.py scripts**
- Present your approach
  - The last lecture consists in project presentations of each group
  - You present the solution you proposed,
    - The methods you applied (data preparation, architecture, training, …)
    - The results you obtained using the train set (eg cross-fold validation results, leave one out, etc, choice up to you)
- Submit your report
  - in practice upload the slides you presented

Task

Dates

- 28.10.2024: Project description
- 13.01.2025: Dedicated 3h slot
- 16.01.2025: Code submission
- 20.01.2025: Project presentations

Thank you for your attention

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